

FNCE 7020 Homework 2

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GitHub: *Repository link to be added.*

Question 1

Table 1 summarizes eight source variables across 161,704 loan relationships, weighting each relationship equally. Firm and bank characteristics therefore enter once per loan. The final row summarizes shock exposure across 41,000 unique firms, counting each firm once. There are no missing values or duplicate loan IDs or firm and bank pairs.

Mean loan growth is -0.061 log points (SD 0.155). Bank assets have a mean of 32.125, a median of 29.220, and an SD of 15.027. The mean preperiod credit share is 0.254, with a median of 0.186. Firm credit quality is centered near zero (mean -0.002 , SD 1.000). Mean firm credit growth and investment growth are -0.016 and -0.005 , with SDs of 0.073 and 0.081, respectively. Across unique firms, shock exposure has a mean of 0.400, a median of 0.353, and an SD of 0.342.

Table 1: Summary Statistics

Variable	Mean	Median	Min	Max	P25	P75	SD	N
Bank assets	32.125	29.220	8.265	103.301	22.303	38.725	15.027	161,704
Supervised	0.465	0.000	0.000	1.000	0.000	1.000	0.499	161,704
Bank shock	0.402	0.000	0.000	1.000	0.000	1.000	0.490	161,704
Pre credit share	0.254	0.186	0.000	1.000	0.078	0.363	0.229	161,704
Firm credit quality	-0.002	0.006	-4.488	4.570	-0.673	0.672	1.000	161,704
Loan growth	-0.061	-0.059	-0.703	0.559	-0.166	0.045	0.155	161,704
Firm credit growth	-0.016	-0.016	-0.342	0.292	-0.065	0.033	0.073	161,704
Investment growth	-0.005	-0.004	-0.356	0.352	-0.060	0.051	0.081	161,704
Shock exposure	0.400	0.353	0.000	1.000	0.053	0.680	0.342	41,000

Notes: Source variables weight loans equally, so firm and bank characteristics repeat across loans. Exposure counts each firm once. Nonmissing IDs identify 161,704 unique loans, 41,000 firms, and 320 banks. $Exposure_f = \sum_b \tilde{w}_{fb} Shock_b$, with $\tilde{w}_{fb} = w_{fb} / \sum_j w_{fj}$, where w_{fb} is the firm's preperiod borrowing share from bank b and $Shock_b$ indicates a funding shock. Normalized shares sum to one within firm. SD is the sample standard deviation. P25 and P75 use linear interpolation. No observations are trimmed.

Table 2 separates sample composition from supervision and shock percentages. Of 41,000 firms, 38,826 (94.698%) borrow from at least two banks and 2,174 (5.302%) borrow from one bank. Panel B reports both the percentage of loan relationships involving a supervised or shocked bank and the percentage of unique banks with that status. Among the 320 banks, 41.875% are shocked and 46.562% supervised. The two columns differ because banks have different numbers of relationships.

Table 2: Sample Composition, Supervision, and Funding Shocks

<i>Panel A: Sample counts</i>		
Loans		161,704
Banks		320
Firms		41,000
Firms borrowing from at least two banks		38,826 (94.698%)
Firms borrowing from one bank		2,174 (5.302%)
<i>Panel B: Supervision and funding shock percentages</i>		
Bank status	Loan relationships (%)	Unique banks (%)
Supervised	46.466	46.562
Bank shock	40.183	41.875

Notes: A loan is a unique firm and bank relationship. Percentages in Panel A use all 41,000 firms as the denominator. In Panel B, the first percentage column uses all 161,704 loan relationships as the denominator and reports the share involving a bank with the stated status. The second uses all 320 unique banks, counting each bank once. Supervised means assets of at least 30; Bank shock means the bank experienced the funding shock. Each status is considered separately.

Question 1: continued

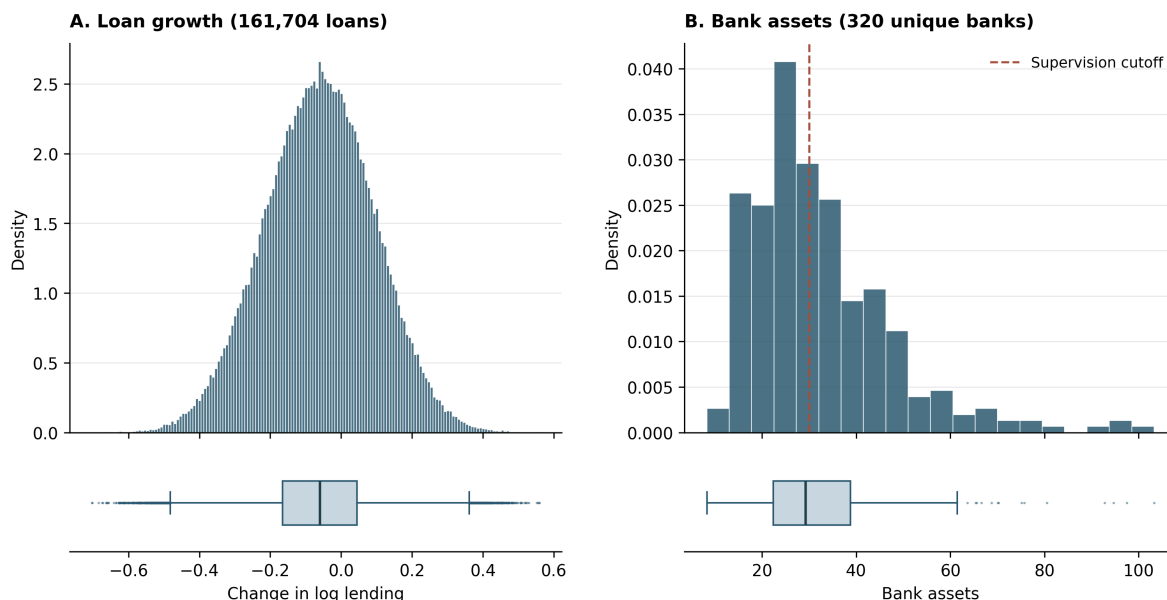


Figure 1: Loan growth across loan relationships and assets across unique banks.

Figure 1 shows that loan growth is approximately symmetric (skewness -0.035) and centered below zero, with a mean of -0.061 and a median of -0.059 log points. The middle 50% of observations lie between -0.166 and 0.045 . Across the 320 unique banks, assets have a pronounced right tail (skewness 1.584): the mean of 32.164 exceeds the median of 29.233 , with a small number of much larger banks. The vertical line marks the supervision cutoff at assets of 30.

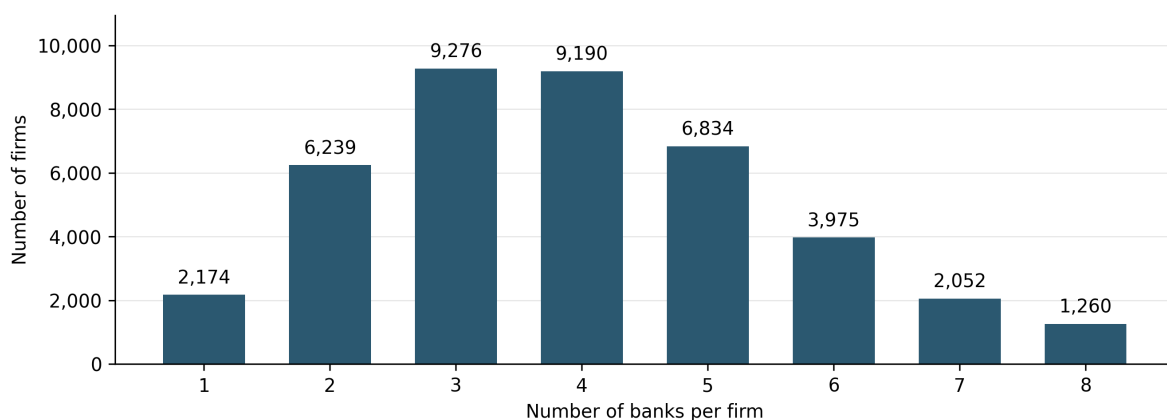


Figure 2: Number of firms by number of bank relationships. Each firm is counted once.

Figure 2 shows that firms borrow from one to eight banks, with a mean of 3.944 and a median of four. Three banks is the most common category, with $9,276$ firms (22.624%). A total of $25,300$ firms (61.707%) borrow from three to five banks, while $2,174$ (5.302%) borrow from only one bank.

Question 2

Design and window sensitivity. Let $\bar{Y}_b$ be the simple mean of loan growth across bank b 's relationships, $x_b = \text{Assets}_b - 30$, and $S_b = 1\{x_b \geq 0\}$. There are 171 banks below the cutoff and 149 above it. Each bank is one observation. Within each window, we fit separate local linear trends on the two sides; the coefficient on supervision is the fitted right limit minus the fitted left limit at 30.

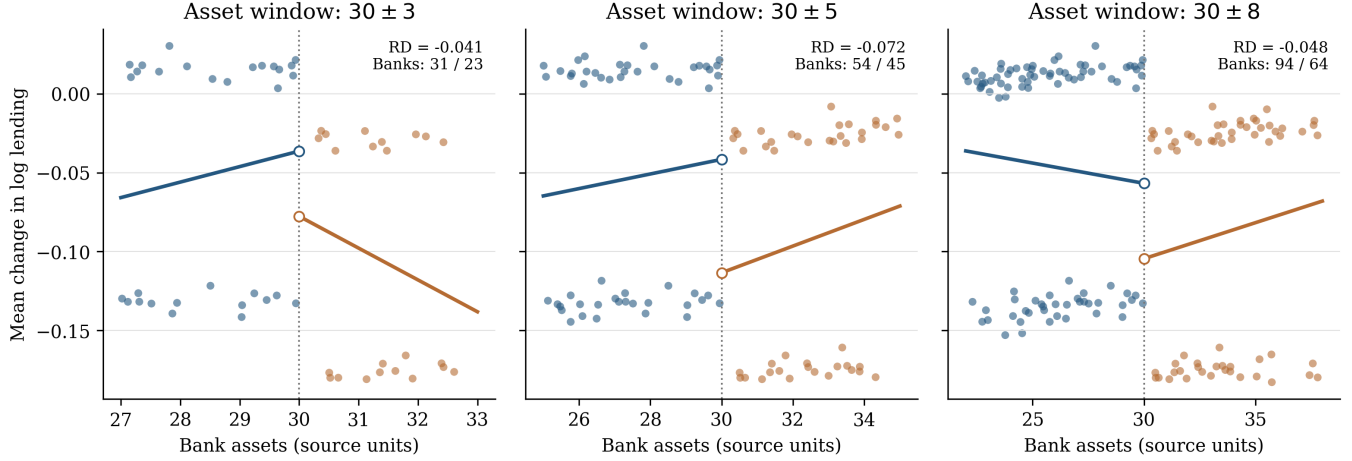


Figure 3: Bank mean loan growth and separate local linear fits for three asset windows.

Table 3: Local Linear Effects of Bank Supervision

Asset window	(1) [27, 33]	(2) [25, 35]	(3) [22, 38]
Supervised	-0.041 (0.043)	-0.072** (0.032)	-0.048* (0.025)
t statistic	-0.959	-2.239	-1.889
p value	0.342	0.027	0.061
Banks below cutoff	31	54	94
Banks above cutoff	23	45	64
Separate asset slopes	Yes	Yes	Yes

Notes: Specification: $\bar{Y}_b = \alpha + \tau S_b + \beta x_b + \gamma S_b x_b + u_b$, where $x_b = \text{Assets}_b - 30$ and $S_b = 1\{\text{Assets}_b \geq 30\}$. The dependent variable is the bank mean change in log lending, kept in source units. Each bank has equal weight within the stated window, equivalent to a uniform kernel. HC3 standard errors are in parentheses. Tests and confidence intervals use a t distribution with $N - 4$ degrees of freedom. These manual estimates do not apply bias correction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3 shows lower bank mean loan growth just above the supervision threshold, with estimated discontinuities between -0.041 and -0.072 log points. The coefficients in columns (2) and (3) are statistically significant at the 5 and 10 percent levels, respectively, whereas the coefficient in column (1) is negative but not statistically significant. The negative sign is stable across windows, although magnitude and significance vary. Wider windows increase precision but rely on comparisons farther from the cutoff.

Table 4: Automatic Bandwidth Selection and Robust RD Inference

	Conventional	Robust bias corrected
RD estimate	-0.057**	-0.062**
Standard error	0.026	0.030
p value	0.025	0.041
95% confidence interval	[-0.108, -0.007]	[-0.122, -0.002]
	Below cutoff	Above cutoff
Estimation bandwidth h	9.776	9.776
Bias bandwidth b	15.087	15.087
Banks within h	107	74
Banks within b	153	97

Notes: The estimand is the jump in bank mean log loan growth at assets of 30. Estimates and standard errors use the source log units. The `rdrobust` specification uses separate local linear fits ($p = 1$), local quadratic bias estimation ($q = 2$), a triangular kernel, the common MSE optimal bandwidth selector `mserd`, and nearest neighbor variance estimation with three neighbors. The right column reports the bias corrected estimate with its robust standard error and normal based confidence interval. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Question 2: continued

Automatic bandwidth and the paper estimate.

My preferred estimate is -0.062 log points, with a 95 percent confidence interval of $[-0.122, -0.002]$, using automatic bandwidth selection and robust bias corrected inference. The estimates across the three fixed bandwidths are also negative, although their magnitude and statistical significance vary. Under the RD assumptions, these results support a negative local effect of supervision on lending growth. However, the confidence interval permits effects close to zero as well as substantially larger declines, and the estimate does not establish effects for banks far from the cutoff.

Validity checks and limits.

Table 5: Limited Balance and Density Diagnostics

Diagnostic	Jump	Std. error	<i>p</i> value
<i>Panel A. Available bank and borrower proxies</i>			
Bank funding shock	16.938	21.602	0.435
Mean borrower credit quality	-0.073	0.109	0.506
Number of loan relationships	-16.227	10.138	0.113
<i>Panel B. Bank asset density</i>			
Density at the cutoff	0.008	0.017	0.659

Notes: Panel A uses the local linear specification in Table 3 within assets $[25, 35]$, with HC3 standard errors and *t* tests. The shock jump is in percentage points, credit quality is in index units, and relationships are counts. These variables are not verified pre supervision covariates: shock timing relative to supervision is unspecified, and borrower composition may respond to supervision. Panel B uses the robust `rddensity` test with local polynomial orders $p = 2$, $q = 3$, a triangular kernel, automatic `comb` bandwidths, and jackknife variance estimation. The density jump is measured per asset unit. All jumps are right minus left. Individual *p* values are unadjusted. Nonrejection does not establish RD validity.

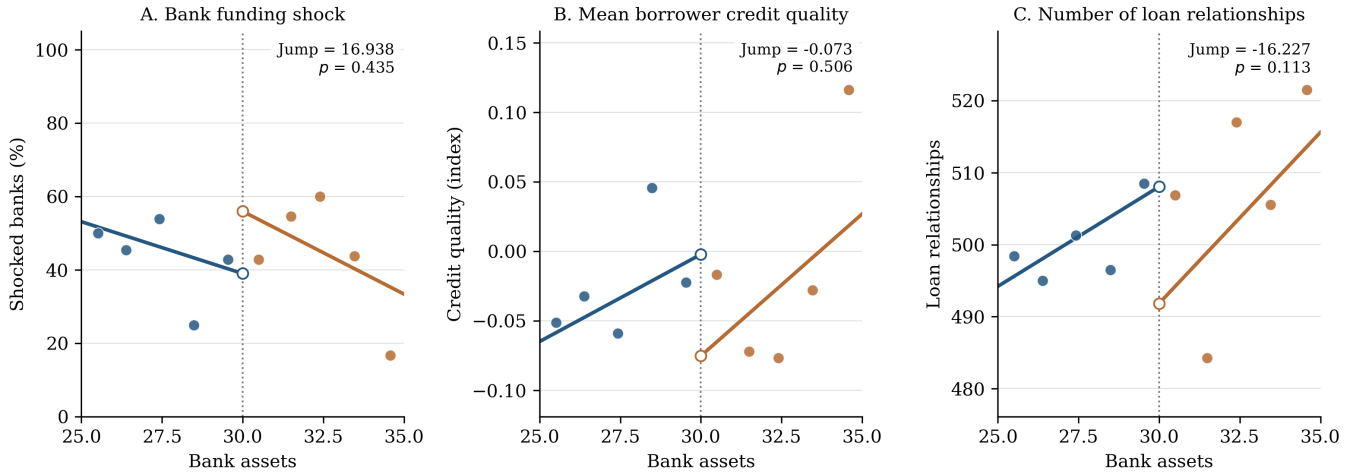


Figure 4: Bank characteristics around the supervision cutoff.

Notes: Filled points show bank means within bins of one asset unit, located at each bin's mean assets. Lines use separate local linear fits to the 99 individual banks in $[25, 35]$, matching Table 5, Panel A. There are 54 banks below 30 and 45 above. Blue denotes the left side and orange the right side. Open circles mark fitted values at 30. The shock jump is in percentage points. These are descriptive checks of the available characteristics, with the timing qualification in Table 5.

Table 5 reports no statistically significant discontinuity in bank funding shock status ($p = 0.435$), mean borrower credit quality ($p = 0.506$), or the number of loan relationships ($p = 0.113$). Figure 4 displays the corresponding binned means and fitted relationships. A discontinuity in a characteristic determined before supervision would suggest sorting or another change at the cutoff, weakening the interpretation of the lending discontinuity as the effect of supervision.

Question 2: continued

Asset density at the cutoff.

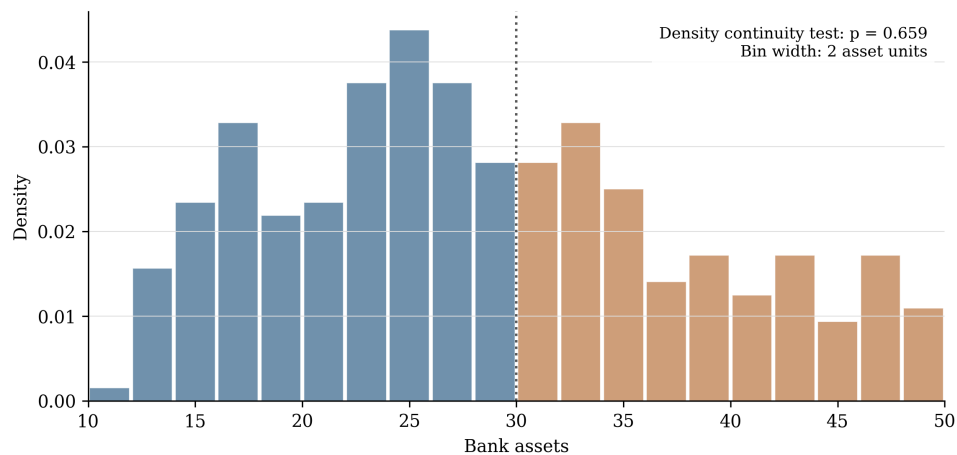


Figure 5: Asset density around 30, displaying assets from 10 to 50. Density is normalized over all 320 banks.

The density test does not reject continuity ($p = 0.659$), so it supplies no statistically significant evidence of bunching just above 30. With only 320 banks and sparse local support, nonrejection is not evidence that manipulation is impossible. A density jump would raise concern about strategic size reporting or selection across the cutoff. Causal interpretation additionally requires continuous potential lending outcomes at 30 and no other intervention changing discretely there. Neither diagnostic proves those conditions.

Question 3

Pooled and within firm comparisons.

Table 6: Funding Shocks and Loan Growth

Sample	(1) Pooled All	(2) Pooled Multiple banks	(3) Pooled All	(4) Firm FE Multiple banks
Bank shock	-0.148*** (0.002)	-0.148*** (0.002)	-0.098*** (0.002)	-0.099*** (0.002)
Credit quality			0.100*** (< 0.001)	
Constant	-0.001 (0.001)	-0.001 (0.001)	-0.021*** (0.001)	
Loans	161,704	159,530	161,704	159,530
Firms	41,000	38,826	41,000	38,826
Banks	320	320	320	320
Firm fixed effects	No	No	No	Yes
R^2	0.220	0.220	0.613	0.194

Notes: Specifications are $Y_{fb} = \alpha + \beta Shock_b + \varepsilon_{fb}$ in columns (1) and (2), with $\gamma Quality_f$ added in (3), and $Y_{fb} = \alpha_f + \beta Shock_b + \varepsilon_{fb}$ in (4). The dependent variable is the change in log lending. Multiple banks means at least two banks per firm. Credit quality is absorbed by firm effects. Parentheses report standard errors clustered by bank and firm. Each cluster component uses its cluster count correction and $(N - 1)/(N - K)$, including absorbed firm effects in K . Tests use 319 degrees of freedom. The last R^2 is within firm. Positive standard errors below 0.001 are shown as < 0.001 ; saved numerical results retain full precision. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6 shows that the pooled shock coefficient is -0.148 , compared with -0.099 with firm fixed effects; both are statistically significant at the 1 percent level. Restricting the pooled sample to multiple bank firms barely changes the estimate, indicating that sample restriction is not the main explanation. Firm effects absorb credit quality and borrower demand, opportunities, and risk shared across lenders. Of 38,826 multiple bank firms, the 29,335 with both shocked and unshocked lenders identify the coefficient.

Direct evidence of confounding.

Table 7: Borrower Credit Quality by Bank Shock Status

	No shock	Shock	Difference	Std. error	p value
Credit quality	0.198	-0.300	-0.498	0.006	< 0.001
Loans	96,727	64,977			

Notes: Means weight each observed loan equally, matching the pooled regression sample. Difference is Shock minus No shock. The standard error is clustered by bank and firm in a regression of credit quality on the shock indicator and a constant. The two sided test uses 319 degrees of freedom. A firm borrowing from both bank types can enter both means.

Table 7 shows that shocked banks lend to borrowers with lower credit quality. The loan weighted means are -0.300 and 0.198 for shocked and unshocked banks, respectively; the difference of -0.498 is statistically significant at the 1 percent level. Controlling for quality reduces the pooled coefficient's magnitude to -0.098 , close to the firm fixed effects estimate. With a quality coefficient of 0.100 , the omitted variable identity for the two pooled regressions gives

$$\hat{\beta}_{\text{short}} - \hat{\beta}_{\text{quality}} = \hat{\gamma} \frac{\widehat{\text{Cov}}(Shock_b, Quality_f)}{\widehat{\text{Var}}(Shock_b)} = \hat{\gamma} (\overline{Quality}_{Shock=1} - \overline{Quality}_{Shock=0}) \approx -0.050.$$

Borrower quality differences therefore make the unadjusted association more negative by approximately 0.050 log points. This decomposition documents sorting and does not establish causality.

Causal interpretation. A causal interpretation of the Khwaja and Mian within firm estimate requires borrower demand and other firm shocks to enter as a common firm component, with bank shock status unrelated to remaining bank or relationship specific determinants of lending:

$$Y_{fb} = \alpha_f + \beta Shock_b + \varepsilon_{fb}, \quad E[\varepsilon_{fb} \mid f, Shock_b] = 0.$$

Under these assumptions, the coefficient identifies the funding shock's credit supply effect for the same borrower. For example, if a firm's shocked lenders are more likely to be supervised and supervision independently reduces lending, firm effects cannot remove this regulatory confounding. Reallocation toward unshocked lenders can also affect the comparison, which does not measure the firm's total credit loss.

Question 4

Exposure and firm outcomes. The analysis uses one observation for each of the 41,000 firms. Exposure measures the share of preperiod borrowing supplied by banks hit by the funding shock:

$$Exposure_f = \sum_{b \in \mathcal{B}_f} \tilde{w}_{fb} Shock_b, \quad \tilde{w}_{fb} = \frac{PreCreditShare_{fb}}{\sum_{j \in \mathcal{B}_f} PreCreditShare_{fj}}.$$

Exposure averages 0.400 and ranges from zero to one. Its coefficient compares full exposure with no exposure. Each firm receives equal weight.

Table 8: Shock Exposure, Total Credit, and Investment

	Total credit growth		Investment growth	
	(1)	(2)	(3)	(4)
Shock exposure	-0.102*** (< 0.001)	-0.040*** (< 0.001)	-0.082*** (0.001)	-0.012*** (< 0.001)
Credit quality		0.060*** (< 0.001)		0.069*** (< 0.001)
Constant	0.024*** (< 0.001)	0.000 (< 0.001)	0.028*** (< 0.001)	0.000 (< 0.001)
Firms	41,000	41,000	41,000	41,000
Credit quality control	No	Yes	No	Yes
R^2	0.226	0.821	0.121	0.745

Notes: Specifications are $Y_f = \alpha + \beta Exposure_f + u_f$ in columns (1) and (3), with $\gamma Quality_f$ added in (2) and (4). Outcomes are total credit growth in (1) and (2) and investment growth in (3) and (4). Exposure is the preperiod borrowing share supplied by shocked banks, with weights normalized within firm to sum to one. Firms are equally weighted. Parentheses report HC3 heteroskedasticity robust standard errors; t tests use $N - K$ degrees of freedom. Inference assumes independence across firms. Positive standard errors below 0.001 are displayed as < 0.001 . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8 shows that greater shock exposure is associated with lower total credit growth and investment growth. Controlling for credit quality reduces the magnitude of the credit coefficient from -0.102 to -0.040 and the investment coefficient from -0.082 to -0.012 . All four exposure coefficients are statistically significant at the 1 percent level. Credit quality is positively associated with both outcomes.

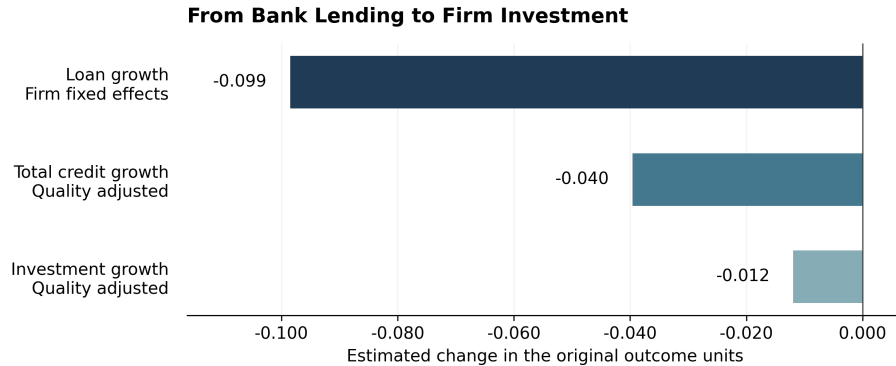


Figure 6: Within firm loan effect and quality adjusted firm effects.

Credit supply transmission. Figure 6 compares the within firm loan estimate (-0.099) with the quality adjusted estimates for total credit (-0.040) and investment (-0.012). The declining magnitudes are consistent with substitution toward other lenders, the use of internal funds, and adjustments to investment timing. These mechanisms are not separately identified. The estimates use different outcome measures and do not imply structural pass through ratios.

Borrower heterogeneity and identification. Lower quality firms have greater exposure to shocked banks, contributing to the larger negative coefficients without controls. Firm effects absorb this borrower heterogeneity in the loan regressions but would also absorb exposure with one observation per firm. Credit quality therefore provides an important control in the firm regressions. A causal interpretation requires exposure to be unrelated to unobserved determinants of credit and investment conditional on quality, with adequate support. Investment opportunities, liquidity constraints, and endogenous lender choice may still generate confounding.